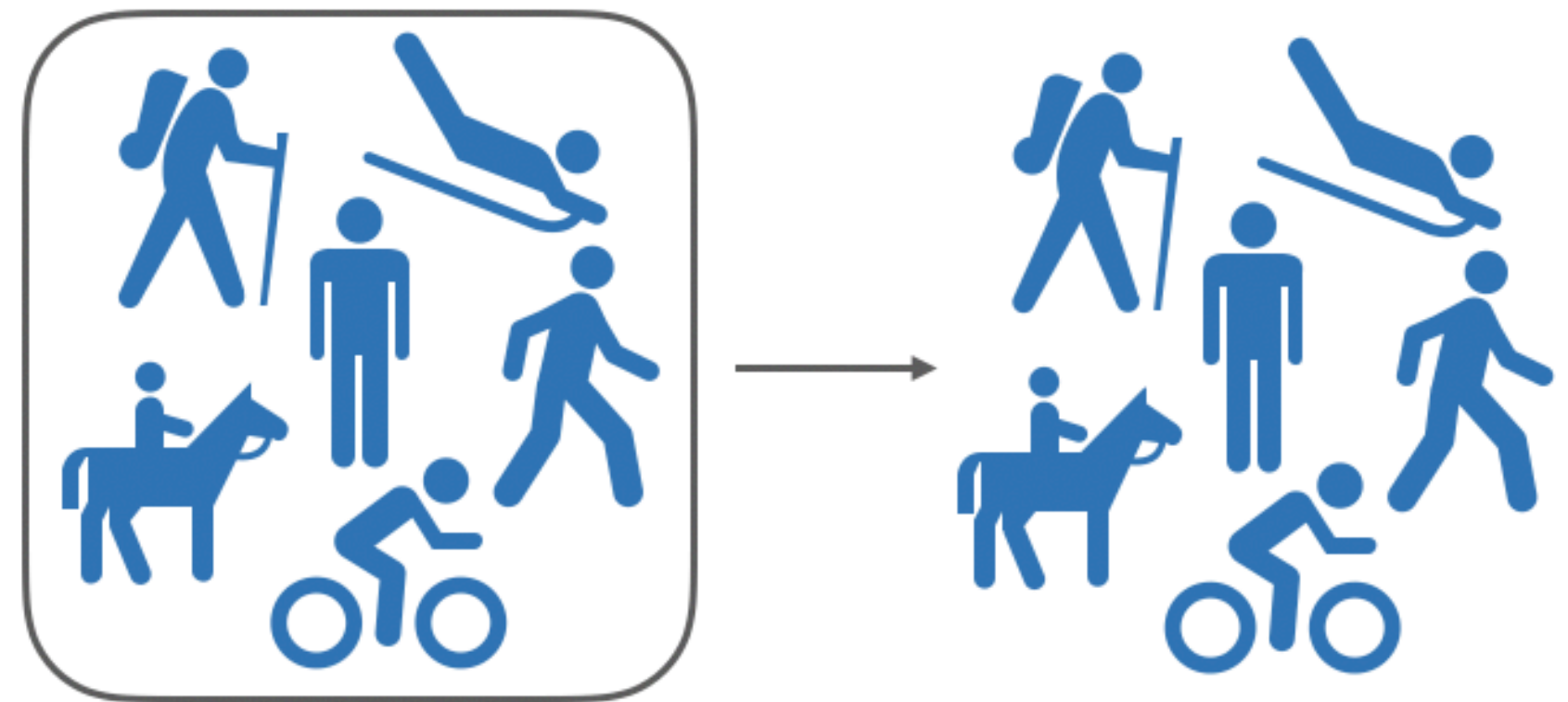


Sparsity-Driven Plasticity in Multi-Task Reinforcement Learning

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Motivation

- ▶ DRL agents lose adaptability over training (neuron dormancy, representational collapse).
- ▶ Achieving SOTA often requires large, inefficient models.
- ▶ At the same time, lacking generalization capabilities observed in other machine learning fields

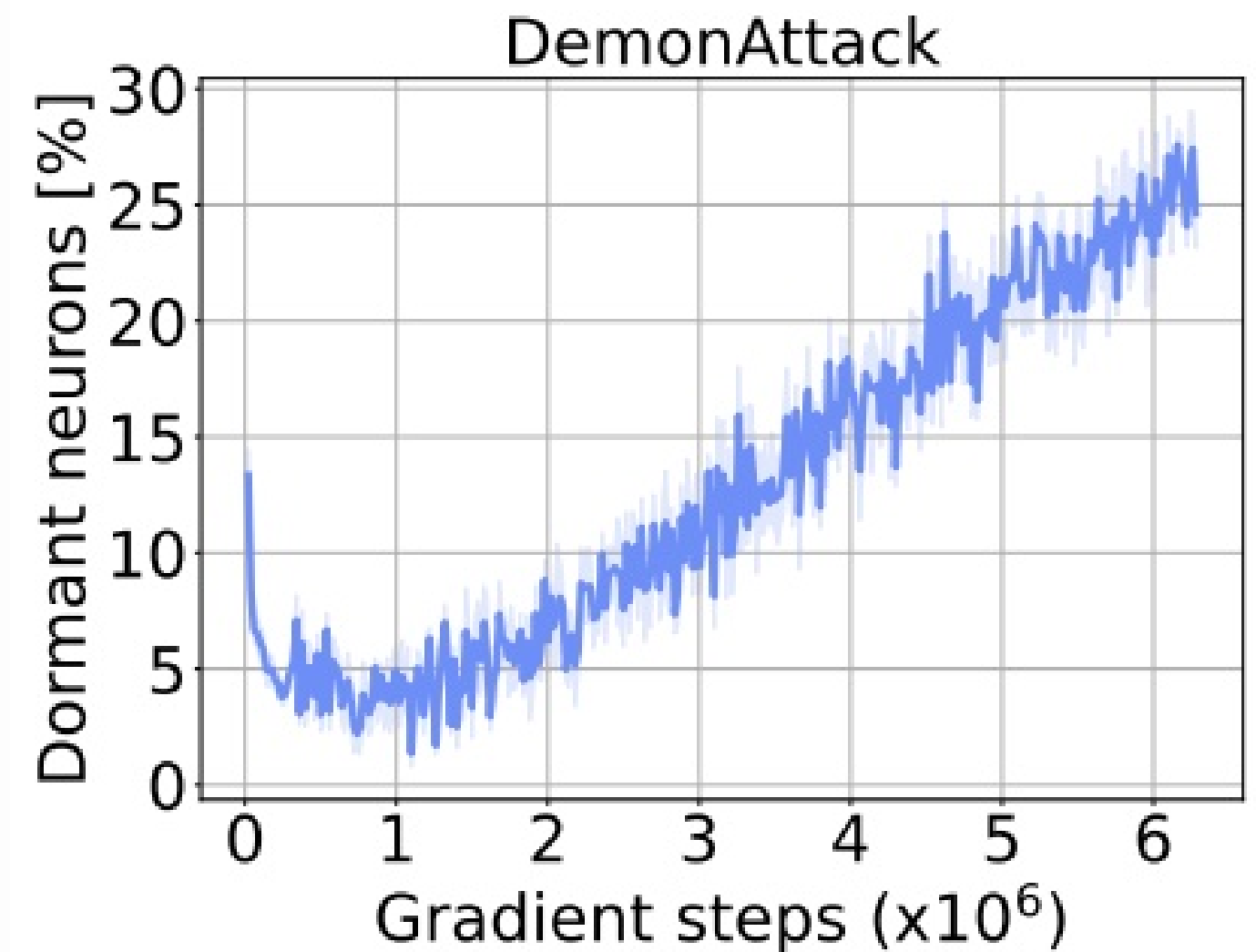
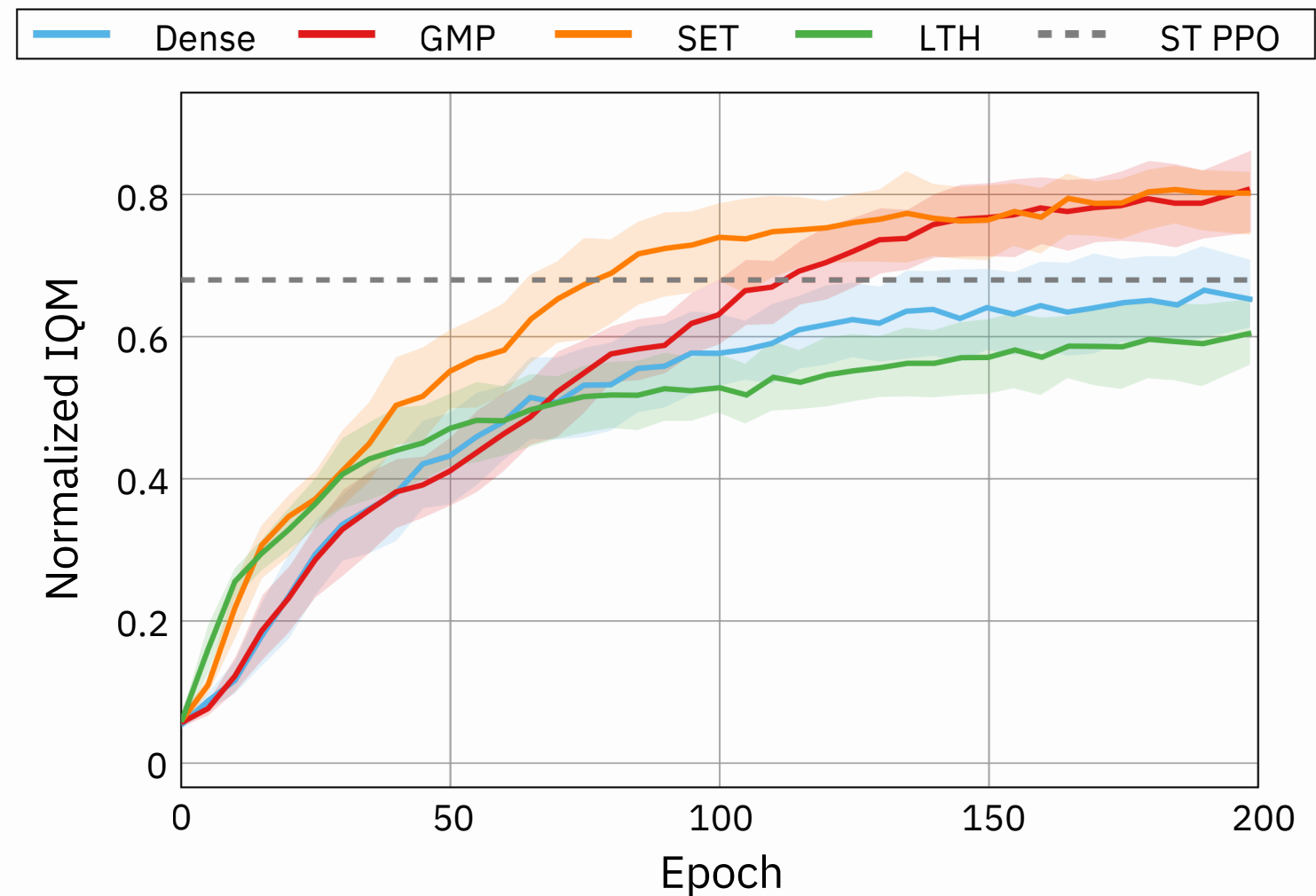


Figure from Sokar et al. (2023). The percentage of dormant neurons of dense networks increases consistently throughout training.

Sparsity is... Effective?



Preliminary experiment on MT5; Methods that adapt their connections *during* training significantly outperform 1) dense, 2) static, and 3) single-task aggregated baselines

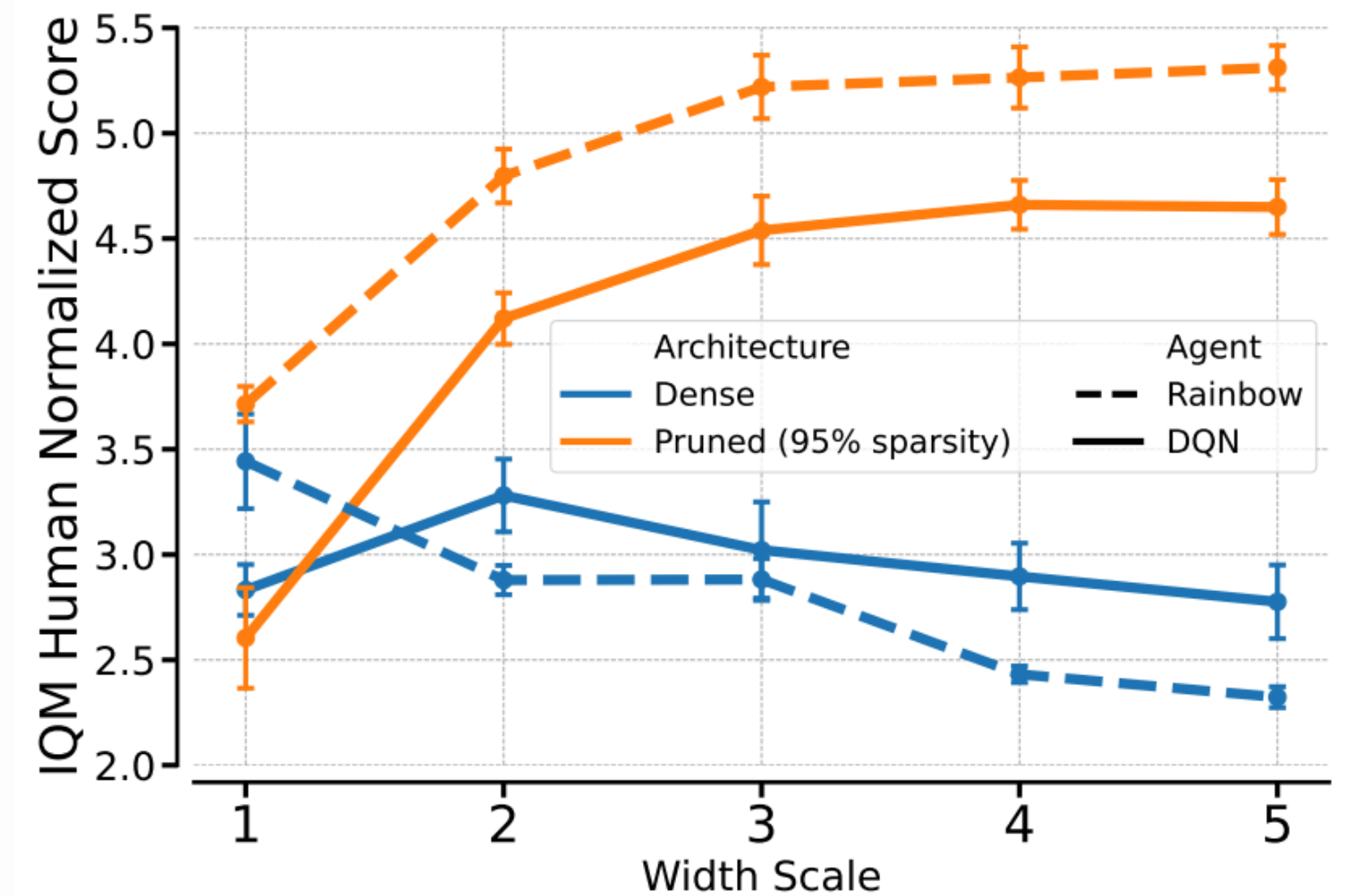


Figure from Obando-Ceron et al. (2024). Pruned networks outperform dense baselines, while being complementary to scaling.

Filling the Gap

MTRL Problem & Context:

1. Requires a higher representational flexibility for diverse and conflicting tasks
2. The persistent challenges haven't been resolved by simple scaling or multi-headed architectures

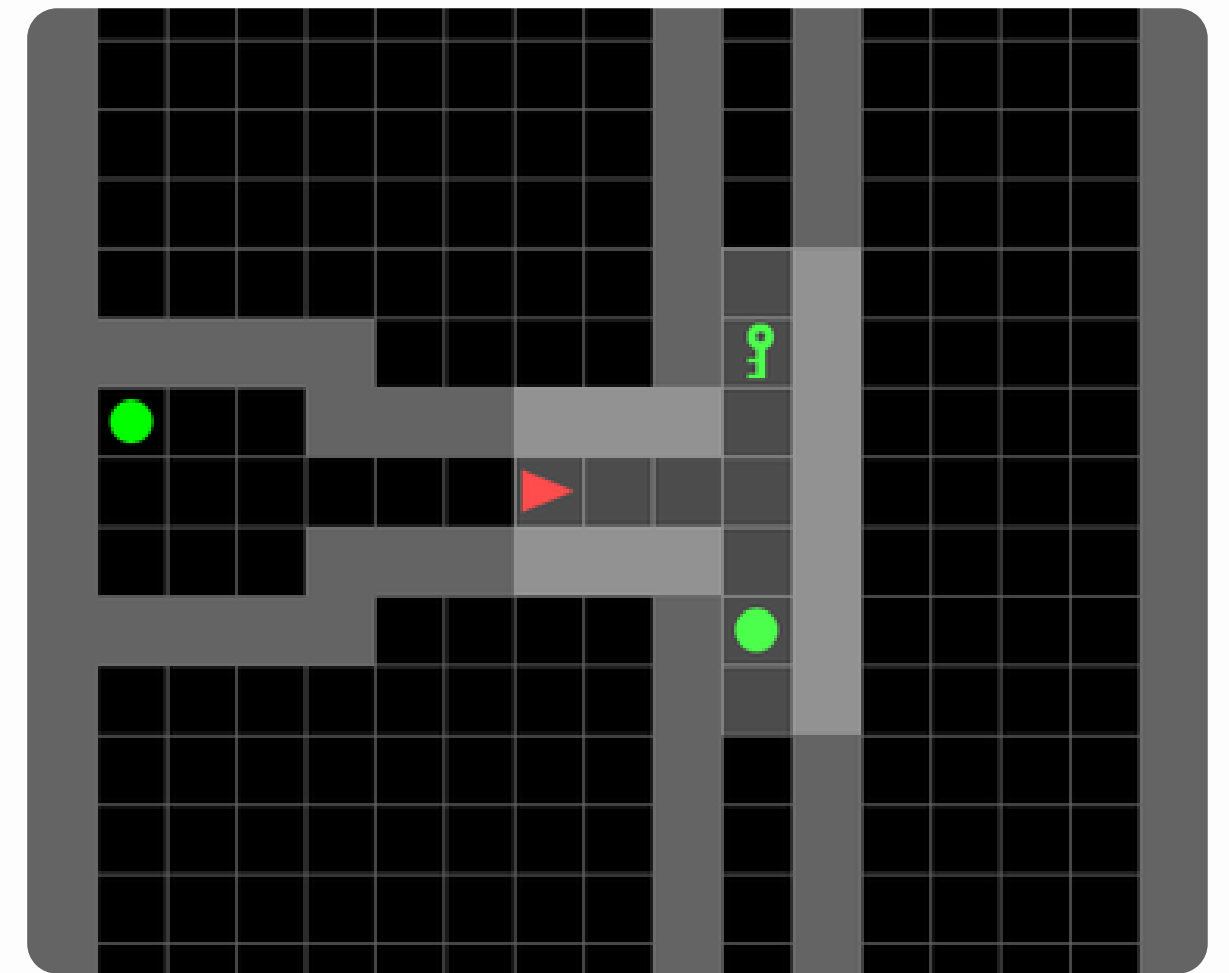
Do sparse methods improve plasticity profiles, thereby increasing performance, in the MTRL setting?

Recent Work & Proposed Avenue:

1. Sparse methods offer solutions that address plasticity pathologies
2. The “bitter lesson” of plasticity

Experimental Setup

- ▶ **MTRL architectures:** Shared Backbone (MTPPO), Mixture of Experts (MoE), Mixture of Orthogonal Experts (MOORE)
- ▶ **Environment:** MiniGrid (MT3, MT5, MT7 benchmarks), partially observable, stochastic objectives, sparse rewards
- ▶ **Plasticity Metrics:** Dormant Neuron %, Effective Rank, FIM Trace

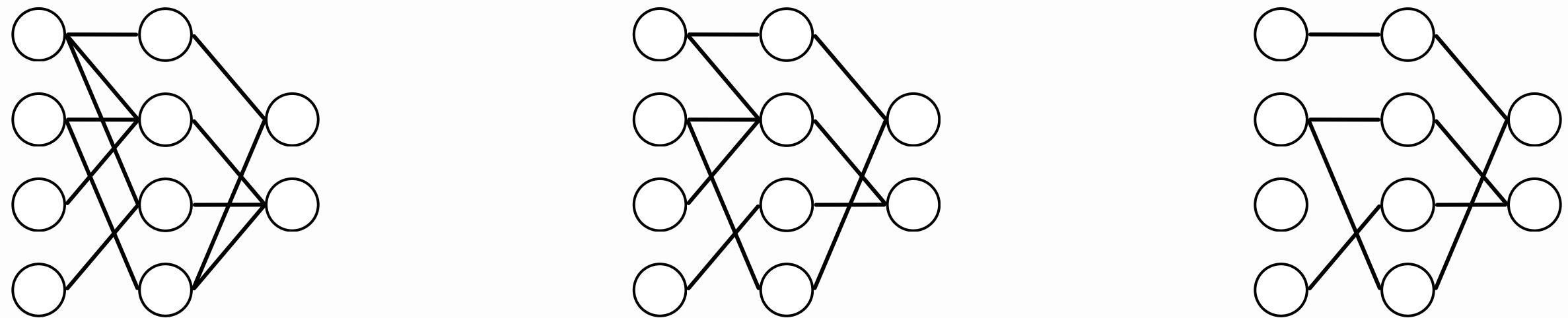


One of the environments used in the benchmarks (Memory). The agent starts in the room to the left, sees a random object, has to walk along the corridor, and go to the object initially seen.

Interventions

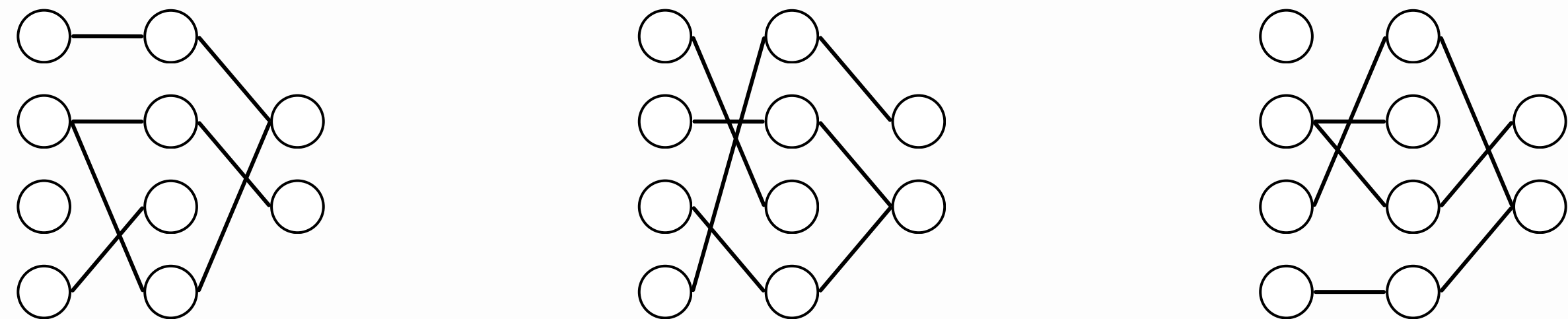
Gradual Magnitude Pruning (Zhu & Ghupta, 2017)

Network becomes more sparse as training progresses



Sparse Evolutionary Training (Mocanu et al., 2018)

Network has a fixed sparsity and disables/regrows connections



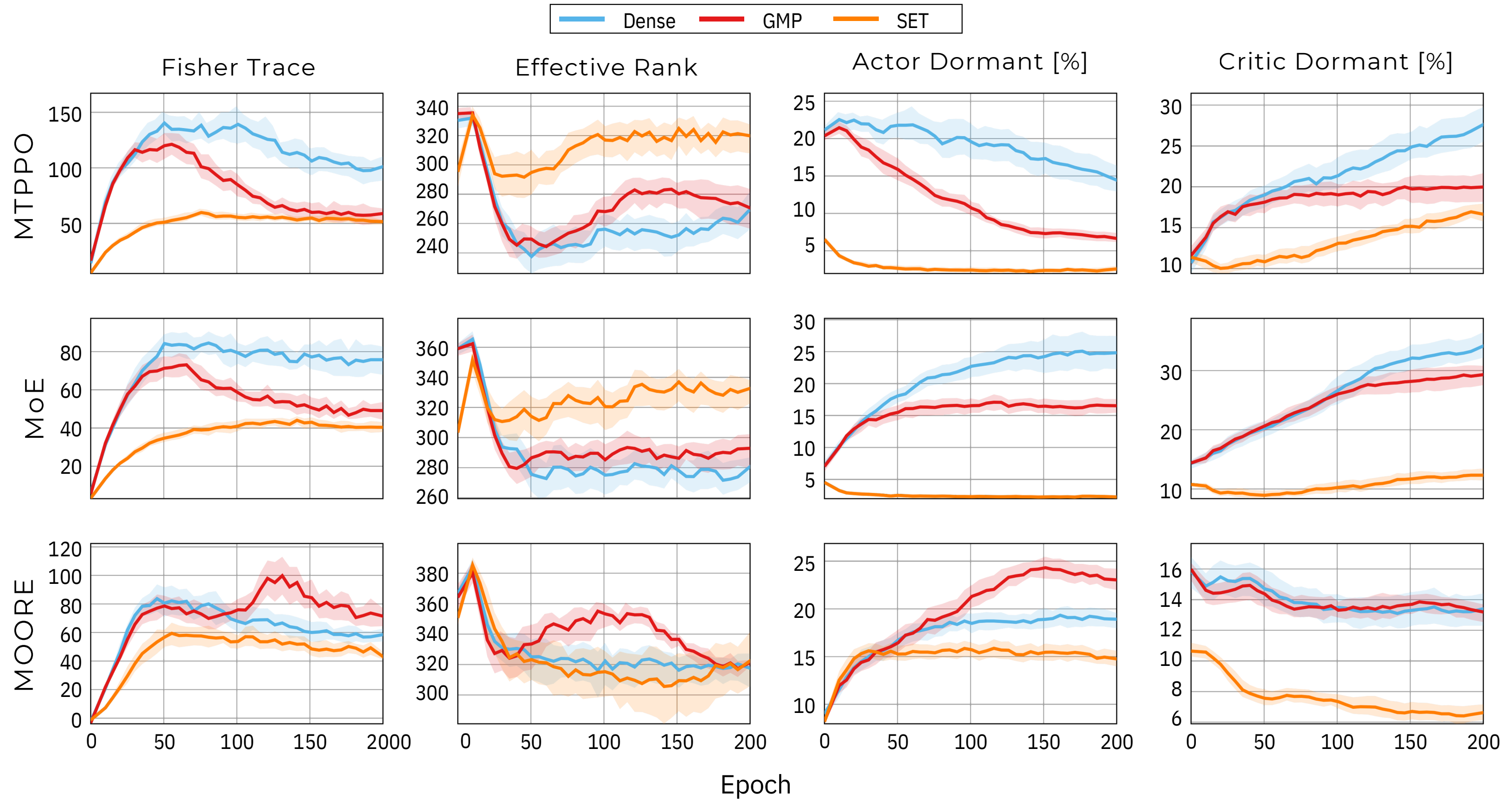
Training steps



Sparse Methods Improve *Performance*

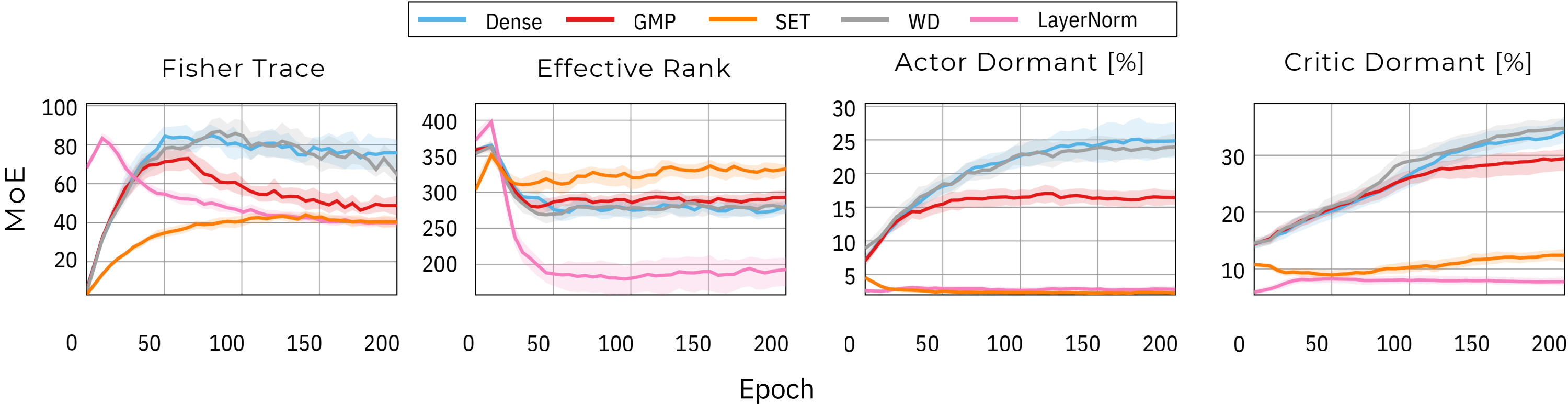
AgentTreatment	MT3		MT5		MT7	
	IQM (↑)	95% CI	IQM (↑)	95% CI	IQM (↑)	95% CI
<i>Multi-Task PPO (MTPPO)</i>						
Dense	0.70	(0.60,0.76)	0.65	(0.61, 0.71)	0.72	(0.69, 0.75)
Gradual Pruning	0.77	(0.73,0.80)	0.81	(0.75, 0.86)	0.76	(0.70, 0.80)
SET	0.76	(0.74,0.77)	0.80	(0.75, 0.84)	0.80	(0.77, 0.84)
<i>Mixture of Experts (MoE)</i>						
Dense	0.74	(0.71,0.76)	0.77	(0.70, 0.82)	0.80	(0.75, 0.84)
Gradual Pruning	0.77	(0.74,0.79)	0.84	(0.78, 0.86)	0.87	(0.83, 0.88)
SET	0.76	(0.74,0.78)	0.79	(0.72, 0.85)	0.82	(0.78, 0.85)
<i>Mixture of Orthogonal Experts (MOORE)</i>						
Dense	0.78	(0.71,0.80)	0.84	(0.81, 0.85)	0.87	(0.84, 0.88)
Gradual Pruning	0.80	(0.77,0.81)	0.85	(0.80, 0.87)	0.88	(0.86, 0.89)
SET	0.69	(0.60,0.74)	0.78	(0.72, 0.83)	0.82	(0.79, 0.85)

Sparse Methods Improve *Plasticity*



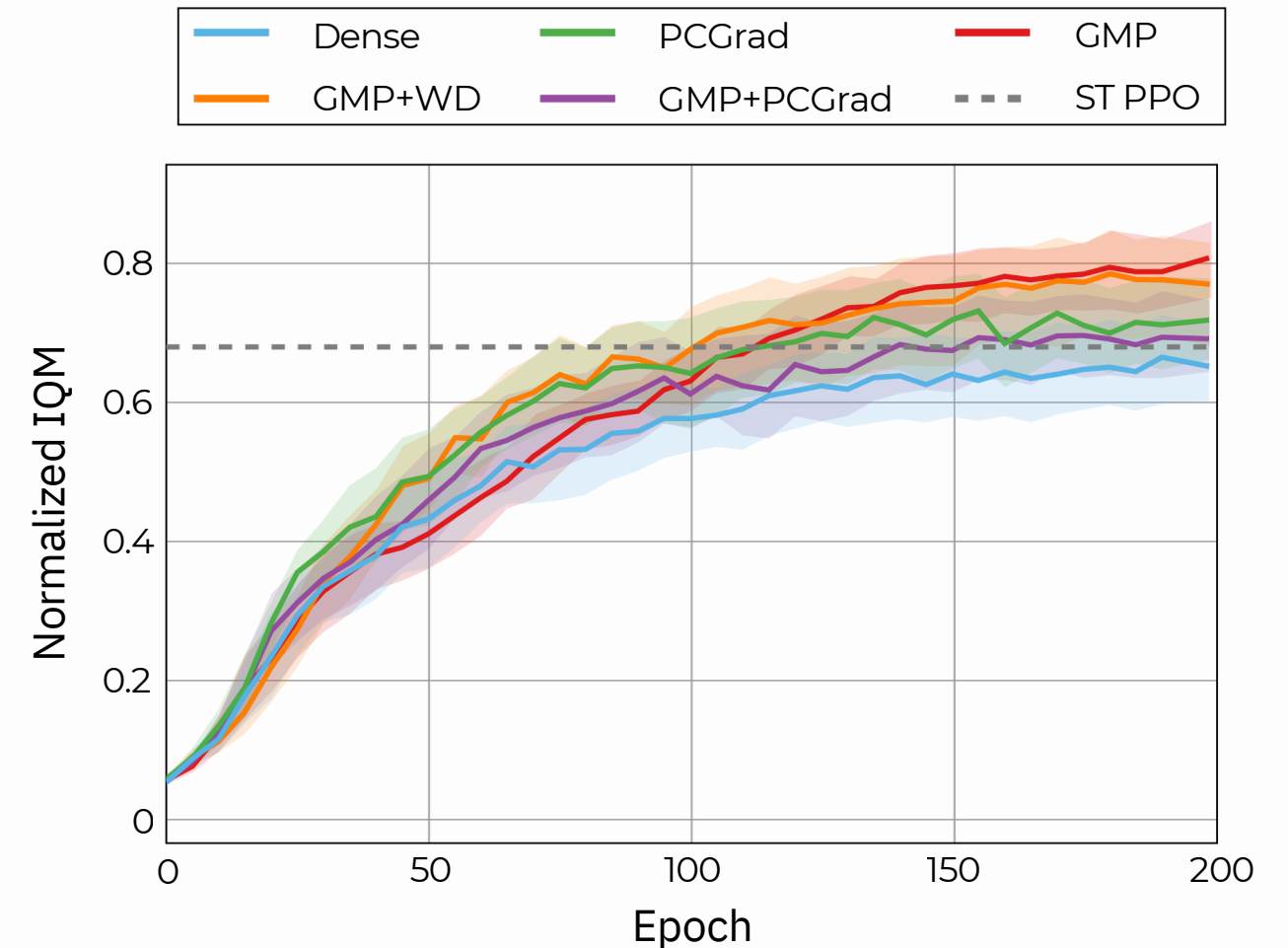
Sparse Methods vs. *Alternatives*

AgentTreatment	MT3		MT5		MT7	
	IQM (↑)	95% CI	IQM (↑)	95% CI	IQM (↑)	95% CI
<i>Mixture of Experts (MoE)</i>						
Dense	0.74	(0.71,0.76)	0.77	(0.70, 0.82)	0.80	(0.75, 0.84)
Gradual Pruning	0.77	(0.74,0.79)	0.84	(0.78, 0.86)	0.87	(0.83, 0.88)
SET	0.76	(0.74,0.78)	0.79	(0.72, 0.85)	0.82	(0.78, 0.85)
ReDo	0.77	(0.76,0.80)	0.82	(0.81, 0.85)	0.85	(0.82, 0.88)
Reset	0.64	(0.54,0.73)	0.78	(0.73, 0.83)	0.84	(0.80, 0.87)
WeightDecay	0.75	(0.71,0.76)	0.78	(0.70, 0.85)	0.77	(0.71, 0.82)
LayerNorm	0.44	(0.38,0.46)	0.34	(0.29, 0.40)	0.39	(0.32, 0.43)



Why & When Sparsity Works

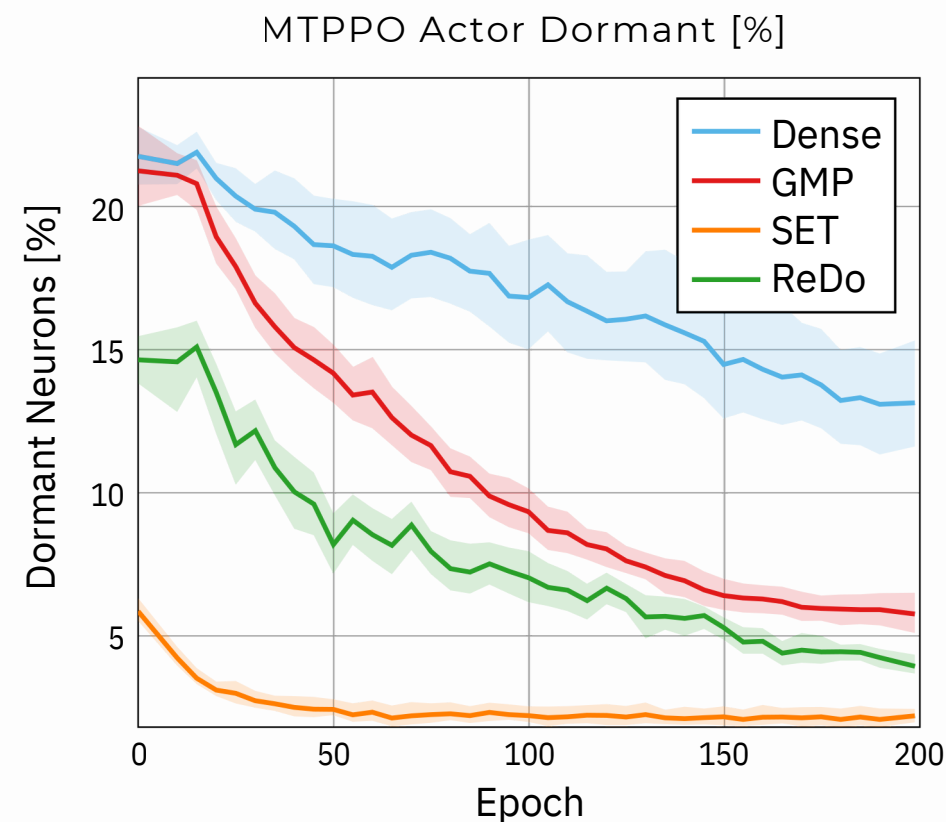
- ▶ Implicit regularization (L_0) → flatter minima → better generalization.
- ▶ Efficient capacity use (less dormancy, more diverse features)
- ▶ The primary benefits of performance & plasticity seem to come from the sparsity itself
- ▶ More nuanced effects when inductive constraints are imposed (MOORE)



Performance of GMP and different synergies with optimizers. While more sophisticated optimizers improve performance over the dense baseline, GMP in isolation achieves the best results.

Conclusion & Limitations

Sparsification is an effective, architecture-sensitive technique for enhancing plasticity and MTRL performance by mitigating plasticity loss indicators.



SET achieves a lower dormant neuron % even compared to ReDo (a targeted method for reinitializing dormant neurons)

Key Takeaway:

Reinforcing the “bitter lesson of plasticity” – simpler, more general techniques are often more beneficial than plasticity-targeting methods.

Limitations:

1. GMP is a computational overhead and does not truly optimize training/inference
2. SET is optimized for fully-connected layers; extensions for recurrent/convolutional layers are not yet developed
3. Hyperparameter tuning still needed (No Free Lunch Theorem). Not one-size-fits-all.

References

- Neil C. Thompson, Kristjan Greenewald, Keeheon Lee, and Gabriel F. Manso. The Computational Limits of Deep Learning, July 2022. URL <http://arxiv.org/abs/2007.05558>. arXiv.
- Ghada Sokar, Rishabh Agarwal, Pablo Samuel Castro, and Utku Evci. The Dormant Neuron Phenomenon in Deep Reinforcement Learning, June 2023. URL <http://arxiv.org/abs/2302.12902>.
- Johan Samir Obando Ceron, Ghada Sokar, Timon Willi, Clare Lyle, Jesse Farebrother, Jakob Nicolaus Foerster, Gintare Karolina Dziugaite, Doina Precup, and Pablo Samuel Castro. Mixtures of Experts Unlock Parameter Scaling for Deep RL. June 2024. URL <https://openreview.net/forum?id=X9VMhfFxwn>.
- Ahmed Hendawy, Jan Peters, and Carlo D'Eramo. Multi-Task Reinforcement Learning with Mixture of Orthogonal Experts, May 2024. URL <http://arxiv.org/abs/2311.11385>.

Not an exhaustive list. For full references, check out the main work :)



Thank you for your multi-headed attention!

Questions?